

Diabetes 101



 PROMEDICA

You just found out you have diabetes. Now what? If you're like most people, you probably have a lot of questions about living with diabetes. We will try to answer some of those questions within this booklet. We also encourage you to talk openly about your condition with your healthcare provider to have your specific questions answered.

Did you know?

Diabetes is a serious chronic health condition that continues to grow across Ohio and Michigan. According to 2009 data by state, it is estimated that 10.1% of Ohio adults and 9.2% of Michigan adults have been diagnosed with diabetes. The Centers for Disease Control and Prevention (CDC) says that in the United States today, 1 in 10 adults has diabetes. And, according to the American Diabetes Association, "If the trend continues, 1 in 3 U.S. adults could have diabetes by the year 2050."



Frequently Asked Questions

What is diabetes?

Diabetes is a disease that causes high blood sugar in your body when it cannot produce enough insulin and/or resists the insulin produced. It is typically a lifelong condition that can affect children and adults. If left untreated, diabetes can lead to other serious medical problems, such as:

- Kidney damage, including kidney failure
- Vision problems, including blindness
- Nerve damage which can cause pain, loss of feeling and digestion problems
- Painful sores and infections on the feet and skin
- Cardiovascular disease

What does insulin do?

Insulin is the hormone that allows sugar/glucose to enter your cells to be used as energy.

What is blood sugar/glucose?

Glucose is the fuel our bodies need for energy. That fuel needs to get into our cells to be used by the body. When someone has diabetes, the glucose stays in the blood stream instead of getting into the cells – starving the cells of energy.

There are three different types of diabetes – **Type 1**, **Type 2** and **Gestational**.

Type 1	Diabetes is a lifelong condition that keeps your body from turning sugar or carbohydrates into energy. When you have type 1 diabetes, your body stops making insulin. Without insulin, your cells can't get fuel to burn for energy. This is why you may feel weak or tired. Managing your diabetes and taking insulin will help you feel better and stay healthy. Type 1 diabetes used to be called juvenile diabetes because it is most commonly diagnosed in children, but it can be diagnosed at any age.
Type 2	Type 2 diabetes is a lifelong condition and is the most common type of diabetes. With type 2 diabetes, the sugar level in your blood is too high. Diabetes keeps your body from turning food into energy. That's why you may feel tired and rundown, especially after eating. Controlling your diabetes means making some changes that may be hard at first. Your healthcare team will help you. Type 2 diabetes is most commonly diagnosed in adults.
Gestational	Gestational diabetes is a special kind of diabetes that happens only during pregnancy. Normally, as food is digested, it turns into sugar (glucose) that goes into your bloodstream. Your body makes a substance called insulin that helps your cells use this blood sugar. Changes that occur in your body while you're pregnant cause your blood sugar to be too high. This can be risky for both you and your baby. You can take steps to control your blood sugar and reduce these risks.

Source: *Krames StayWell*



Why do I have to measure my blood sugar/glucose?

Blood sugar testing provides you with vital information on how to manage your condition. Knowing your glucose level can help you:

- Reach your treatment goals
- Understand how your daily choices in diet and exercise or stress levels affect your condition
- Understand how your medication is working for you

How do I measure my blood sugar/glucose levels?

Blood glucose is measured by taking a drop of blood and putting it on a strip. The meter is able to tell you what your blood sugar is at the time you test it. Sometimes, your doctor will have you test it multiple times a day.

What could happen if I don't know my blood sugar/glucose levels?

Most of the time people do not know if their blood sugar is elevated unless it is very high or very low.

Some symptoms of high blood sugar are:

- Extreme thirst
- Frequent urination
- Dry skin
- Blurred vision
- Drowsiness

Some symptoms of low blood sugar are:

- Shaking
- Sweating
- Dizziness
- Hunger
- Impaired vision
- Weakness
- Fatigue
- Irritability
- Loss of consciousness

Blood sugar fluctuates throughout the day, so it is a good idea to check your bloods sugar levels at various times during the day to be sure your blood sugar is under control.

What is the A1C test?

The A1C test is a blood test that shows what your blood sugar levels have been, on average, for the past three months.

What does food choice have to do with diabetes?

Carbohydrates are our major source of energy and have the greatest effect on blood sugar. High-fiber carbohydrate foods like whole grain bread, whole fruits, legumes, nuts, and oats digest slower so they help prevent elevations of the blood sugar.

What does exercise have to do with diabetes?

Exercise helps insulin work more efficiently and can help keep blood sugar in a normal range.

Is there a medicine I can take for diabetes?

If you have type 1 diabetes, insulin must be taken every day because your body does not make its own insulin. With type 2 diabetes, there are a wide variety of oral medications that can be used to help control your diabetes.



Strategies for Managing your Diabetes

Our doctors and diabetes educators have put together a list of strategies you may be able to use to help you cope with your diabetes. These include:

- Following your meal plan
- Exercising regularly
- Learning to recognize signs of high and low blood sugar
- Wearing medical identification
- Using your support systems
- Talking about your feelings
- Learning how to solve problems
- Thinking positively and act assertively
- Practicing stress management
- Seeing your doctor regularly
- Continuing to learn about your diabetes

Setting up a Diabetes Tool Kit

Your diabetes should not get in the way of spending time away from home. You may want to consider making a tool kit to help you manage your condition while you are away from home. Some items you may want to keep in your toolkit are:

- Extra insulin, syringes, pens, or insulin pump supplies
- Other medications you take for diabetes-related problems
- Copies of all prescriptions
- The pharmacy label that came with your insulin (this is needed for air travel)
- Fast-acting sugar, such as glucose tablets, for hypoglycemia
- Glucagon for severe hypoglycemia
- A blood glucose meter, lancets, test strips, and a log book
- Extra batteries for your meter
- An ID card that says you have diabetes and lists emergency contact numbers

Source: *Krames StayWell*



Diabetes Education

Once you've met with your doctor to understand your new diagnosis, you should plan to meet with one of our certified diabetes educators. Diabetes educators will explain how you can live your life fully with diabetes. They can assist with a plan for your treatment and show you how to monitor your blood glucose levels. You also will learn how to plan meals and exercise in ways that help keep your diabetes under control – and allow you to enjoy your life without worrying about your health.

Your certified diabetes educator will take time to understand what you know about your condition and what you are doing to manage it today. You'll learn about what diabetes is, how it affects you, when you should eat and take your medication, when to test your blood and what the meter is telling you. You and your diabetes educator will work together to set goals for your ongoing care.

After your first meeting, there are a series of classes you will attend in our Diabetes Self-Management Training program.

Overview Class

Learn how the body uses sugar, what your lab results mean, what are high and low blood sugar values, and what monitoring guidelines you should use.

Problem Prevention Class

Learn how to prevent complications from high and low blood sugar, such as increased risk of heart disease, stroke, vision loss, kidney failure, sexual issues, foot care, and nerve damage.

Exercise Class

Learn the benefits of exercise, its effect on blood sugar, safety issues, stress management, weight loss, and how to fit exercise into your lifestyle.

Stress Management Class

Learn the effect of stress on blood sugar, and identify daily stress, good and bad stress, and management techniques.

Final Follow-Up Class

Find out how well you are controlling your blood sugar, review your goals and progress, and review strategies to keep your blood sugar under control. This may be a good time to ask any additional questions you may have about diabetes.

Medical Nutrition Therapy

Your doctor also may feel you would benefit from nutrition counseling like our Medical Nutrition Therapy program. This program offers a unique approach to nutrition and healthier eating through four separate stages.

Initial Assessment

First, you will meet with a dietitian and learn what a serving of carbohydrates is and how to balance them throughout the day. You will also learn how to read labels, choose appropriate snacks, eat healthy when dining out, and how alcohol affects blood sugar levels. Your dietitian may also suggest cookbooks, web sites, apps, and recipes that can help you manage your diabetes.

Follow-up Appointment

At your follow-up appointment, you will learn what to eat if you are sick, how to limit sodium intake, which fats are healthier and how to monitor fat intake, and the effect of fiber on blood sugar control. This appointment will also focus on learning which foods are high in fiber and which foods are nutrient rich.

Six-month Group Class

In this small group setting, you'll learn to feel comfortable controlling your blood sugar. In this class, we help you evaluate goals, review what you are eating, check your blood sugar levels, and answer any questions or concerns.

Annual Follow Up

After completing these steps, you should meet with the nurse to review any questions you have regarding your diabetes control and/or the dietitian to review your meal plan and eating habits. You will need to get a new order from your doctor to do this.

Exercise for Diabetes

When managing your diabetes, exercise is a key in controlling your blood sugar/glucose levels. You'll learn about how to exercise safely from your diabetes educator. We also recommend that you talk with your doctor about what exercise program is right for you.

Medication Therapy Management

At The Pharmacy Counter, we have found that if you understand your medication and how you will benefit by taking it, you may have better outcomes. So, we have developed a Medication Therapy Management (MTM) program where you to meet with one of our pharmacists and review your medications (prescriptions, supplements and over-the-counter medications). We discuss how and when is best to take them. At your appointment, the pharmacist will work with you to be sure you understand the benefits of your medication. We also coordinate with your primary care physician should you require some changes in your medications to improve your results.



Managing your Diabetes

As a road map to keeping you healthy, we recommend the following plan of care:

Every Three Months	• Visit your primary care doctor • Have an A1C test (if blood sugar/ glucose is not stable) • Check your weight and blood pressure, and check your feet
Every Six Months	• Have an A1C test • See your dentist for an exam and cleaning
Every Year	• See your primary care doctor for a full physical exam • Receive a comprehensive foot exam • Have lipid profile, kidney test, serum creatinine test, and flu vaccine • Visit an ophthalmologist or optometrist for a dilated eye exam • Visit with a dietitian for medical nutritional therapy (MNT) • Meet with a pharmacist to review your medication (MTM) • Visit with a nurse for diabetes self-management training (DSMT)

About ProMedica's Diabetes Services

At the ProMedica Mary Ellen Falzone Diabetes Center, our Mission is to improve the health and well-being of those with diabetes.

We are honored that the Falzone family has chosen to help others in our community with their gift in support of a comprehensive diabetes center. Mary Ellen Falzone (Meme) was very much a typical 14-year-old girl until she died unexpectedly in 1978 from complications with diabetes – a disease she and her family did not even know she had.

Today, Meme's legacy lives on through the new Center, which will help provide patients with the tools and encouragement they need to manage their disease and live full lives.

Some of the services provided at the Center will include:

- Comprehensive diabetes education classes
- Early treatment and education for newly diagnosed patients
- Home care diagnostic and glucose monitoring evaluation
- Insulin pump classes
- Training for home glucose monitoring
- Specialized care, including telehealth programs for uncontrolled Type 1 diabetes
- Nutrition services
- Medication Therapy Management

ProMedica's diabetes services extend to all of the communities we serve through satellite programs across northwest Ohio and southeast Michigan.



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www.promedica.org/diabetes